Installation Guide

Version 1.0.1 · rechecked 2026-11-01.

See the Thai illustrated installation guide (INSTALLATION_TH.md) and validation report (VALIDATION.md). This remains a public preview: mock tests and schema/compiler checks are not a successful live tenant or Desktop smoke test.

1. What this package does

Copilot Token Dashboard is a tenant-owned Power BI reference application. It provides ten report pages for personal usage, organization aggregates, agent operations, models, value scenarios, budgets and data quality.

It does not grant access to another organization's data. Every recipient uses their own Microsoft tenant, Power BI workspace, source permissions and secrets. The installation demonstration is synthetic and is marked on every page.

Illustrated help: see the ten-page walkthrough in section 11 or the sample dashboard gallery (SAMPLE_DASHBOARD.md). You can explore the live synthetic sample (https://zangkasi55.github.io/Copilot-Token-Dashboard/) or the offline sample dashboard (assets/sample-dashboard.html) without a tenant or sign-in: download the HTML file and open it in Edge or Chrome.

2. Prerequisites

Component	Required for
Windows and recent Power BI Desktop	Local report authoring and validation
Python 3.10 or later	Collectors, normalized-data build and deployment
PowerShell 7	Installation script
Power BI Pro, PPU or suitable Fabric capacity	Publishing and sharing, as required by Microsoft licensing
Workspace Contributor or higher	Provisioning semantic model and report
Azure CLI, or a tenant-scoped Fabric token in your process environment	Automated tenant provisioning
Approved source permissions	Collecting actual organizational usage and billing data

The supplied API origins target Microsoft's global commercial cloud. Sovereign clouds and custom GitHub hosts are not validated by this release.

Power BI Desktop and optional build utilities are not redistributed.

3. Download and validate

Download the versioned installation ZIP from the GitHub Releases page. Verify its SHA-256 checksum against the release checksum file, then extract it into a new folder, for example C:\CopilotTokenDashboard.

```
Set-Location C:\CopilotTokenDashboard
.\Install.ps1 -Action Validate -InstallDependencies
```

This restores only the declared runtime dependencies and validates the complete synthetic table set. No tenant resources are created in Validate mode.

Expected result: the validation reports all contract tables and no errors.

Do not disable enterprise execution policies or security controls to run the script. Follow your organization's approved script-execution process.

4. Open the demonstration locally

```
.\Install.ps1 -Action Local -SourceMode Demo
```

The script creates a new version-specific folder under your local application data and opens the PBIP. An existing installation folder is not overwritten. Use -InstallFolder C:\CTD\Demo-1.0.1 to choose a new folder.

In Power BI Desktop:

1. Wait for the project to open.
2. Select Refresh if the data is not yet loaded.
3. Confirm the synthetic-demo banner and data-through date.
4. Visit all ten report pages.
5. Use the selector on Agent Detail to review a single agent.
6. Open Modeling > View as and test the Viewer role with an example UPN.

The example people use the reserved example.com domain. The sample prices are fictional; they are not vendor prices or a recommended rate card.

5. Publish manually into your own tenant

This path does not require an automated deployment token.

1. Sign in to Power BI Desktop with an account in the destination tenant.
2. Create or select an approved Power BI workspace.
3. Publish the opened project from Power BI Desktop.
4. Configure credentials and refresh in the Power BI Service.
5. Assign approved Entra security groups to the semantic model's Viewer role.
6. Test as real Viewer users before sharing the report or publishing an app.

A PBIT, when supplied, is opened in Desktop and instantiated there. It is not directly imported through the Power BI Service REST API. PBIP projects are editable source projects, not executable copies of Power BI Desktop.

6. Automated tenant provisioning

Use an explicitly chosen tenant and workspace:

```
.\Install.ps1 -Action Tenant -SourceMode Demo `
  -TenantId '<your-tenant-guid>' `
  -WorkspaceId '<your-workspace-guid>'
```

Azure CLI authentication is performed against the specified tenant if no CTD_FABRIC_ACCESS_TOKEN is present. Alternatively, acquire an approved tenant-scoped token with SemanticModel.ReadWrite.All and Report.ReadWrite.All, or Item.ReadWrite.All, and set it in the current process environment.

The script checks the token's tenant before sending any provisioning request. The API still verifies the token's signature, audience and permissions.

The installer creates the semantic model first, then binds the report to the created model ID. It follows Fabric long-running operations and does not silently overwrite an existing item. Use -Update only after reviewing the destination and backing up its current definitions.

The resulting state is provisioned, configuration required. Security-group assignment, data-source credentials, refresh, app audiences and Viewer smoke tests remain required. The script does not falsely declare production readiness.

7. Replace synthetic data with your own data

Never edit the shipped demonstration to contain real secrets or employee data and then commit it. Work in an ignored runtime folder.

```
python -m pip install .
ctd collect-github --settings config\settings.json `
  --day 2026-05-28 --output runtime\incoming\github.jsonl
ctd collect-m365 --settings config\settings.json `
  --output runtime\incoming\m365.jsonl
```

Copy config\settings.example.json to the ignored config\settings.json and provide your tenant/app/source settings. Secrets are read from environment variables or from an approved secret manager that populates those variables.

Prepare the input folder:

- People.csv, Scopes.csv, ScopeAccess.csv: required, reviewed organization

and access mappings.

- usage.jsonl: normalized metadata records, combining the approved sources.
- Models.csv, Surfaces.csv, Agents.csv, Budgets.csv, Outcomes.csv: optional

registry and outcome inputs using the exact public contract.

- period-summaries.jsonl: optional Microsoft 365 period summaries.

Use the data contract and source-capability guide. Do not spread period totals into daily rows or infer cached tokens from input totals.

```
ctd build-data --input runtime\incoming --output runtime\gold-2026-05-28
ctd validate-data --input runtime\gold-2026-05-28 --production
```

The build sets demonstration mode to false for customer data, removes ineligible organization scopes, erases suppressed numeric facts and preserves unpriced/unattributed records.

8. Customer data-source modes

CSV

```
.\Install.ps1 -Action Local -SourceMode CSV `
  -DataFolder C:\CTD\runtime\gold-2026-05-28 `
  -InstallFolder C:\CTD\Customer-1.0.1
```

The Power BI Service cannot read your local folder without a configured on-premises data gateway. A Desktop-published snapshot can be viewed without scheduled refresh, but it is not a live pipeline.

SharePoint

Upload a complete, validated generation of the contract CSV files to an approved folder in your own tenant. Use the exact folder URL, including its trailing slash, and configure organizational OAuth

credentials:

```
.\Install.ps1 -Action Tenant -SourceMode SharePoint `
  -TenantId '<your-tenant-guid>' -WorkspaceId '<your-workspace-guid>' `
  -DataFolder C:\CTD\runtime\gold-2026-05-28 `
  -SharePointSite 'https://<your-tenant>.sharepoint.com/sites/AIReporting' `
  -SharePointFolder 'https://<your-tenant>.sharepoint.com/sites/AIReporting/Shared Documents/Gold/'
```

The local validated folder is required as a preflight mirror. The model reads the SharePoint files after deployment, not that local mirror.

A file must match its expected name and exact folder once; missing or duplicate matches fail explicitly. Credentials are not shipped in the model.

Version 1.0.1 validates that the HTTPS folder is genuinely under the selected site and has its trailing slash before copying/deployment. Local CSV paths are resolved to absolute paths. Data validation also rejects malformed Settings, incorrect row widths, unknown fact references and small unsuppressed cohorts.

9. Security and go-live checks

- Use the Viewer RLS role; app page visibility is not a security boundary.
- Give consumers Viewer/app access, not Workspace Member or Contributor.
- Do not grant broad Build/export permissions without a data-protection review.
- Assign your own sensitivity label in the destination tenant.
- Verify personal user A cannot see user B's personal rows.
- Verify a small scope is absent and suppressed aggregate rows contain no

numeric facts, even through dataset access.

- Configure refresh only after a complete data generation is uploaded.
- Test all source credentials and permissions in the destination tenant.
- Confirm no demo identities remain in customer data.
- Review access and source coverage periodically.

This is not a legal-compliance certification or an employee-performance tool.

10. Troubleshooting

Symptom	Action
Python dependency error	Install the declared runtime package; do not install arbitrary packages
CSV header/type error	Compare exact column order/types with the contract

Missing tokens or cached tokens	Inspect source coverage; do not substitute zero
Unexpected spend totals	Compare billed/estimated basis, unattributed rows and privacy-hidden breakdowns
No personal rows	Confirm the actual UPN mapping and Viewer role membership
Blank organization scope	Confirm approved scope access and the six-person minimum
HTTP 403	Obtain required consent/role/policy; do not bypass the restriction
Existing item conflict	Inspect the destination; use Update only deliberately
Refresh failure	Configure credentials, gateway and source-folder permissions
Long-running operation timeout	Inspect the workspace and operation before retrying creation

The accompanying administrator and source-capability manuals provide detailed operating boundaries.

11. Illustrated sample dashboard walkthrough

The following screenshots were captured from the standalone browser sample using the shipped synthetic CSV data. They are illustrations, not verified Power BI Desktop screenshots. The companion uses a simplified layout and different styling; it does not run DAX, call APIs, test authentication or enforce RLS. Production acceptance still requires the actual Power BI project in your tenant.

For figures 1–6 and 8–10, select Last 28 days and Example engineering team. Figure 7 uses Example operations team. The personal page uses the fixed fictional viewer user01@example.com. All examples end on 2026-05-28. Click an image on GitHub to view its original 1600-pixel-wide PNG; the PDF embeds the image so it is available offline and can be zoomed.

Figure 1. Overview: check the scope and data date

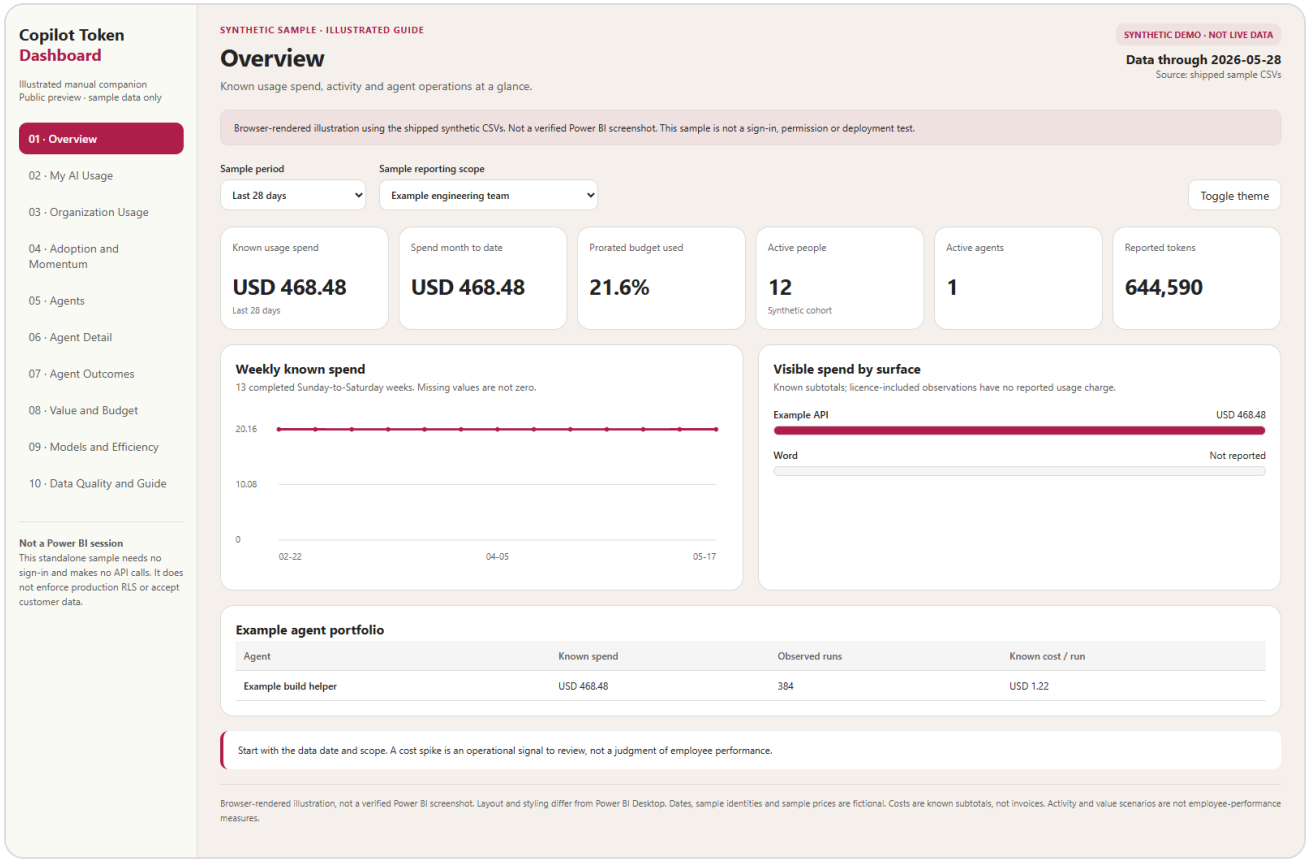


Figure 1. Browser-rendered synthetic Overview illustration, not a verified Power BI screenshot.

1. Check the SYNTHETIC DEMO banner and data-through date before reading a value.
2. For the engineering example, known 28-day spend is USD 468.48, with

12 active people, 1 active agent and 644,590 reported tokens.

3. The weekly chart covers completed weeks. The current spend spike is not

retroactively added to earlier completed weeks. Word's missing usage price remains unavailable instead of being treated as a metered zero.

Figure 2. My AI Usage: validate the fictional self-view

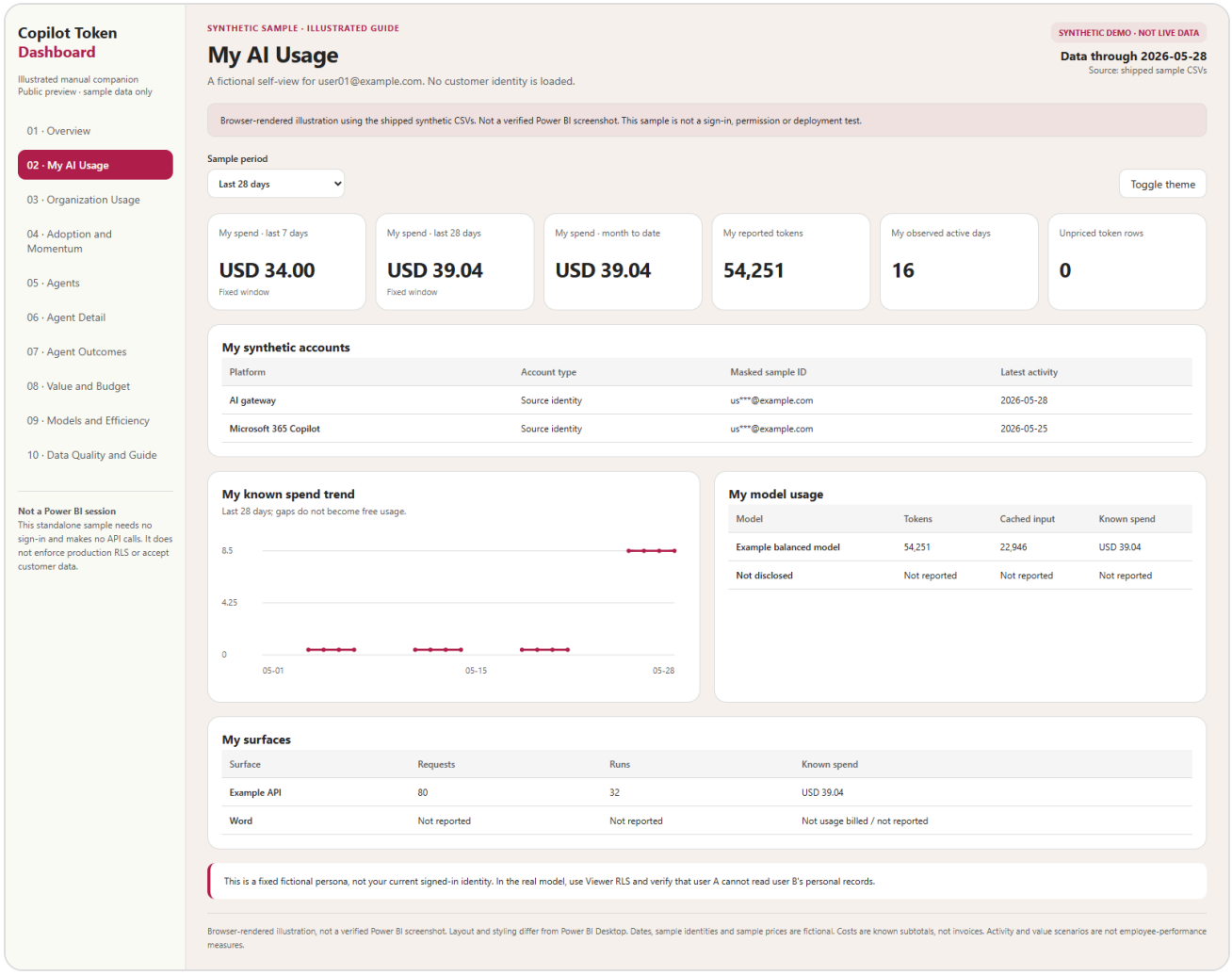


Figure 2. Browser-rendered synthetic personal usage illustration for user01@example.com, not a verified Power BI screenshot.

The example user's spend is USD 34.00 for seven days, USD 39.04 for 28 days and USD 39.04 month to date. The tables show masked accounts, reported model tokens and unavailable fields explicitly.

In the real Power BI model, test the Viewer role with two actual UPNs. Never use this fixed browser persona as evidence that tenant RLS works.

Figure 3. Organization Usage: understand privacy and completed weeks

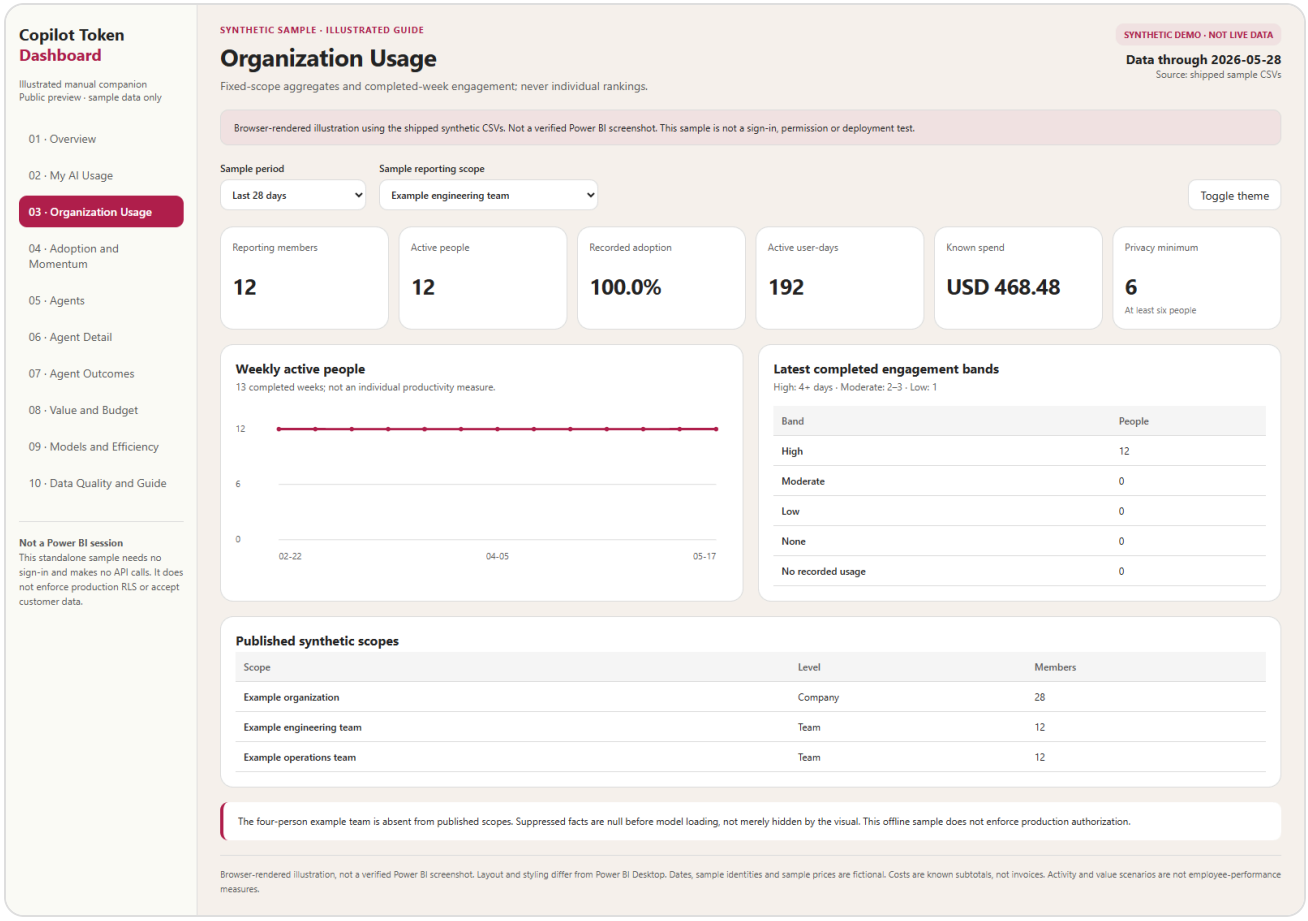


Figure 3. Browser-rendered synthetic organization usage illustration, not a verified Power BI screenshot.

The engineering scope contains 12 members. A four-person example team is absent from the published scope list because the minimum is six. High engagement means four or more observed days in a completed week; it is not a productivity rating. The HTML has no authorization boundary; production protection comes from the reviewed aggregate pipeline and the Power BI Viewer role.

Figure 4. Adoption and Momentum: interpret the two growth axes

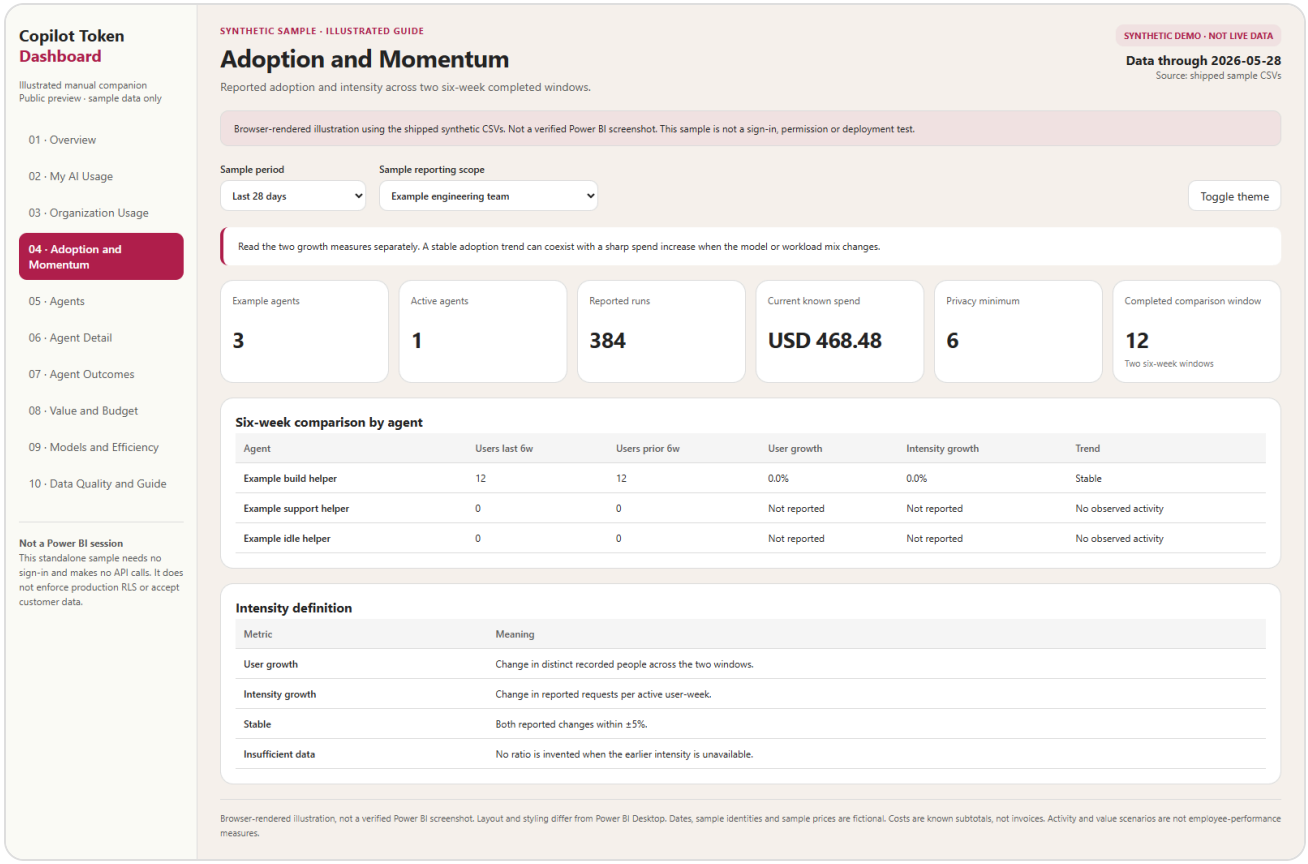


Figure 4. Browser-rendered synthetic adoption and momentum illustration, not a verified Power BI screenshot.

Compare the latest six completed weeks with the preceding six. User growth and requests per active user-week are different measures. The example build helper is Stable in these windows even though its latest spend spikes, because the spike falls outside the last completed comparison window.

Figure 5. Agents: investigate a spend flag

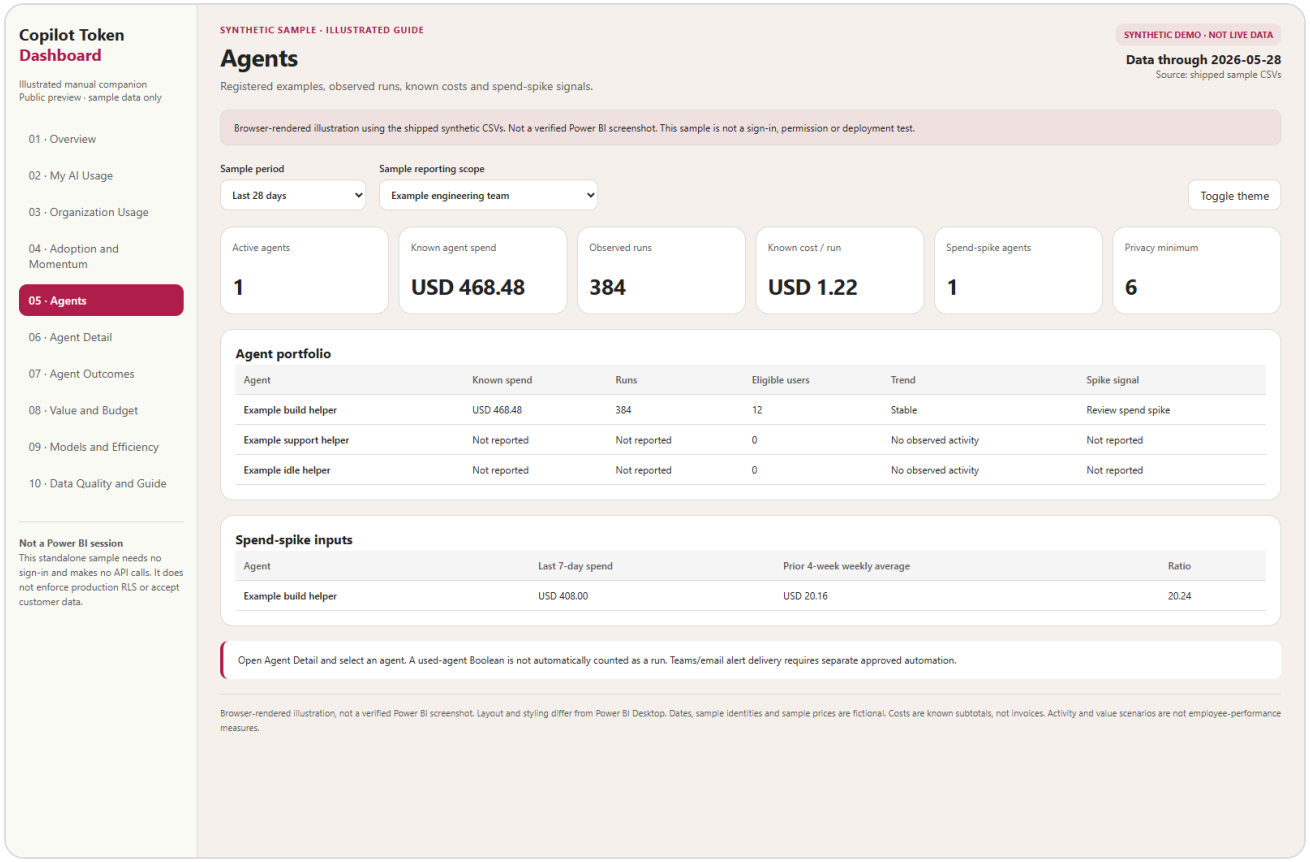


Figure 5. Browser-rendered synthetic agent portfolio and spend-spike illustration, not a verified Power BI screenshot.

The example build helper has 384 reported runs in the selected period. Its last-seven-day spend is USD 408.00, compared with a prior four-week weekly average of USD 20.16. Review the workload and source coverage instead of interpreting the flag as an employee-performance judgment. Notifications require separate approved automation.

Figure 6. Agent Detail: select one registered example

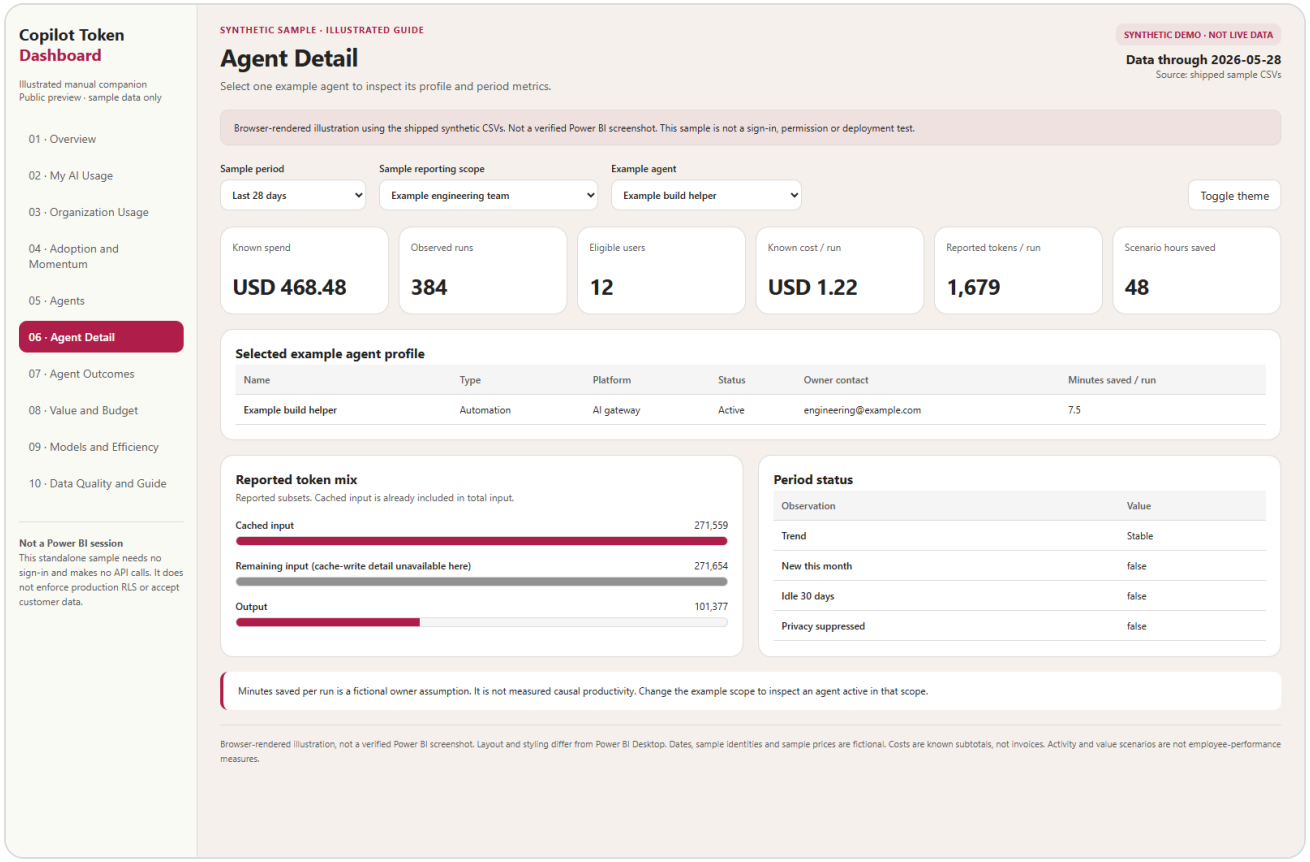


Figure 6. Browser-rendered synthetic agent detail illustration, not a verified Power BI screenshot.

Open Agent Detail and use the agent selector. The build helper's known cost per run is approximately USD 1.22. Its seven-and-a-half-minute saved-per-run setting is a fictional owner assumption, not a measured benefit.

If an agent has no activity in the current scope, change to its approved scope. Do not replace absent metrics with invented values.

Figure 7. Agent Outcomes: use the operations example

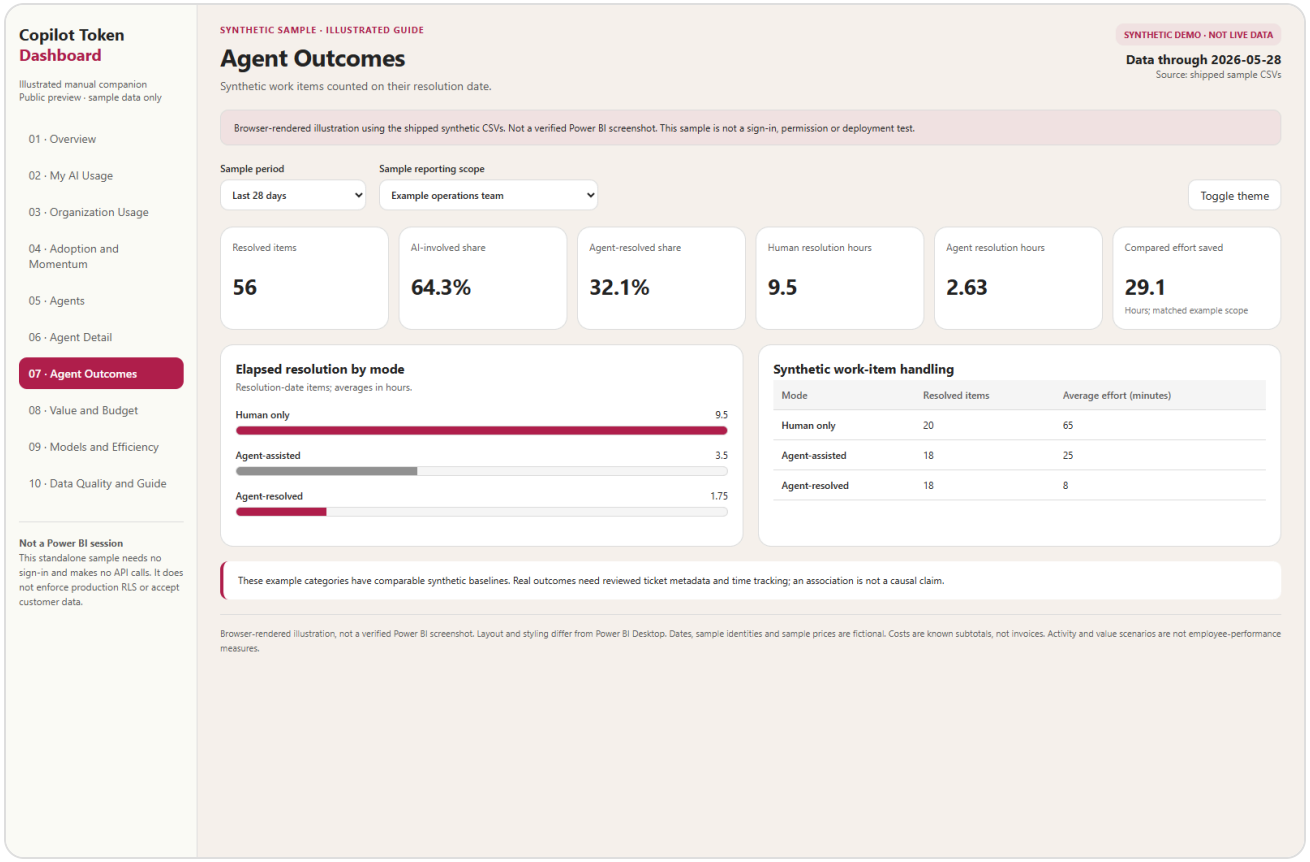


Figure 7. Browser-rendered synthetic work-outcome illustration for the operations team, not a verified Power BI screenshot.

Select Example operations team. The sample has 56 resolved items: 20 human-only, 18 agent-assisted and 18 agent-resolved. Elapsed hours and tracked human effort are separate quantities. The example comparison does not establish causation or replace review of your own ITSM data.

Figure 8. Value and Budget: test assumptions, not guarantees

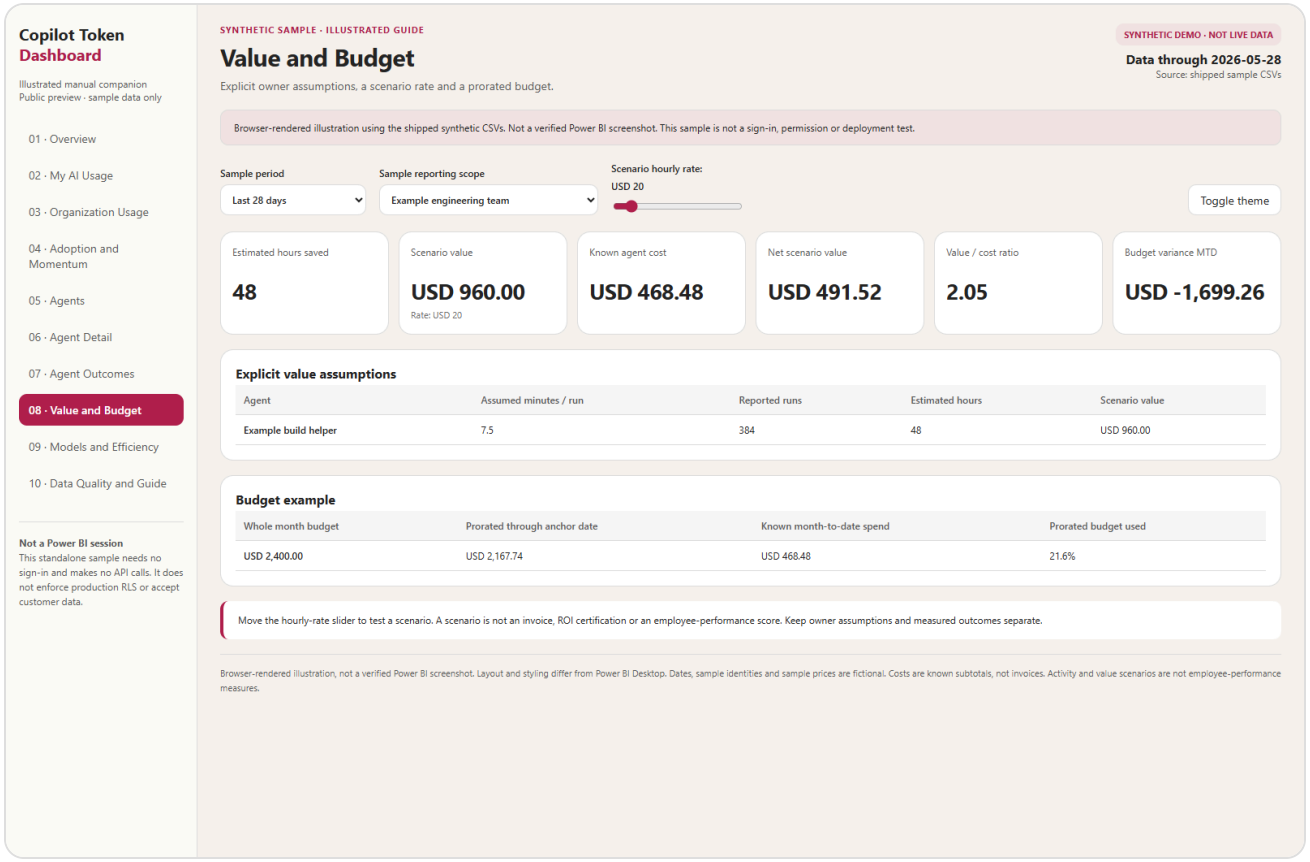


Figure 8. Browser-rendered synthetic value and budget scenario illustration, not a verified Power BI screenshot.

At USD 20 per hour, the build-helper example shows 48 estimated hours and USD 960.00 scenario value. Changing the sample slider to USD 40 doubles the value to USD 1,920.00; it does not change measured spend or prove productivity. Budget is prorated through the displayed anchor date.

Figure 9. Models and Efficiency: reconcile known subtotals

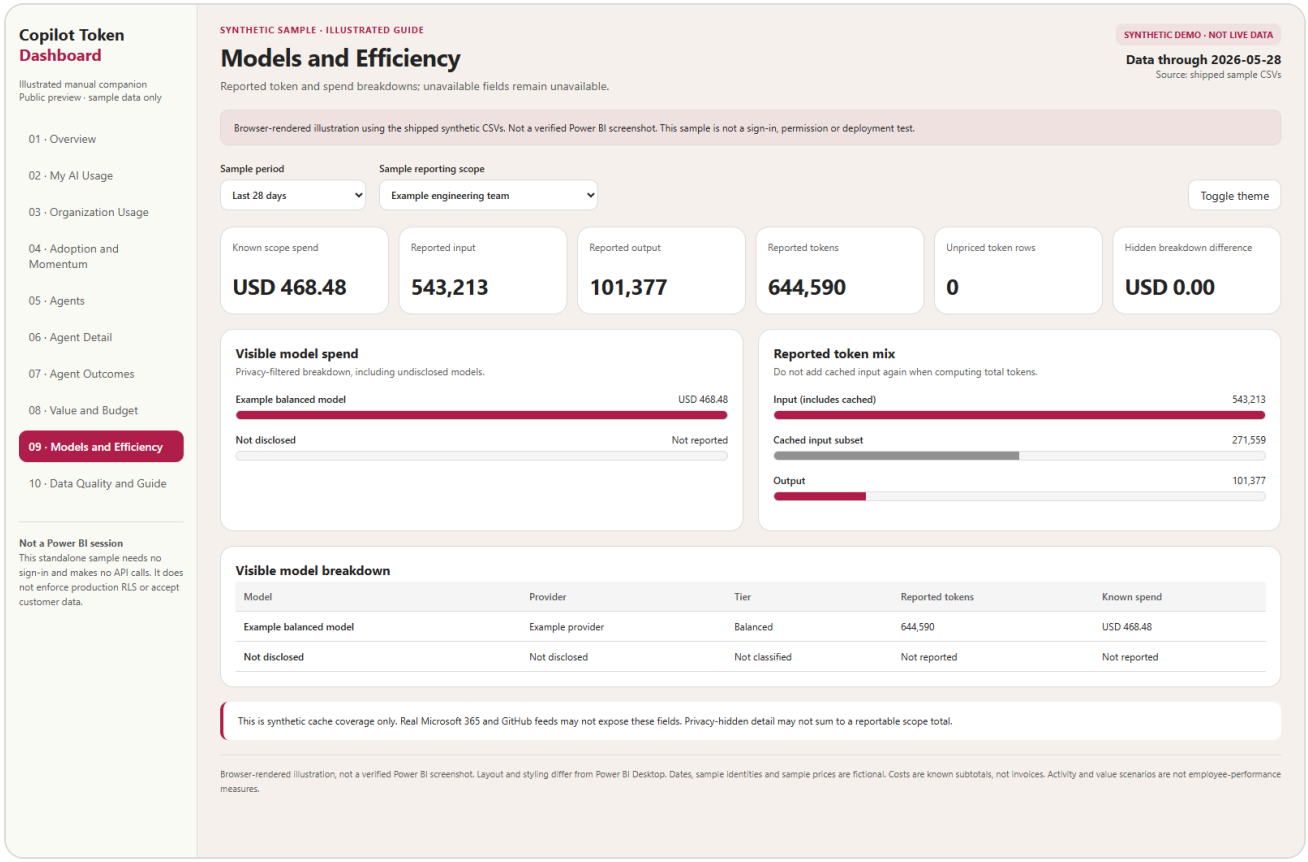


Figure 9. Browser-rendered synthetic model and token-efficiency illustration, not a verified Power BI screenshot.

Input includes the cached-input subset: do not add cached tokens a second time. Compare known model/surface subtotals with the scope total and inspect any privacy-hidden difference. The sample's fictional cache coverage is not a promise that Microsoft 365 or GitHub APIs expose the same fields in production.

Figure 10. Data Quality and Guide: finish the installation checklist

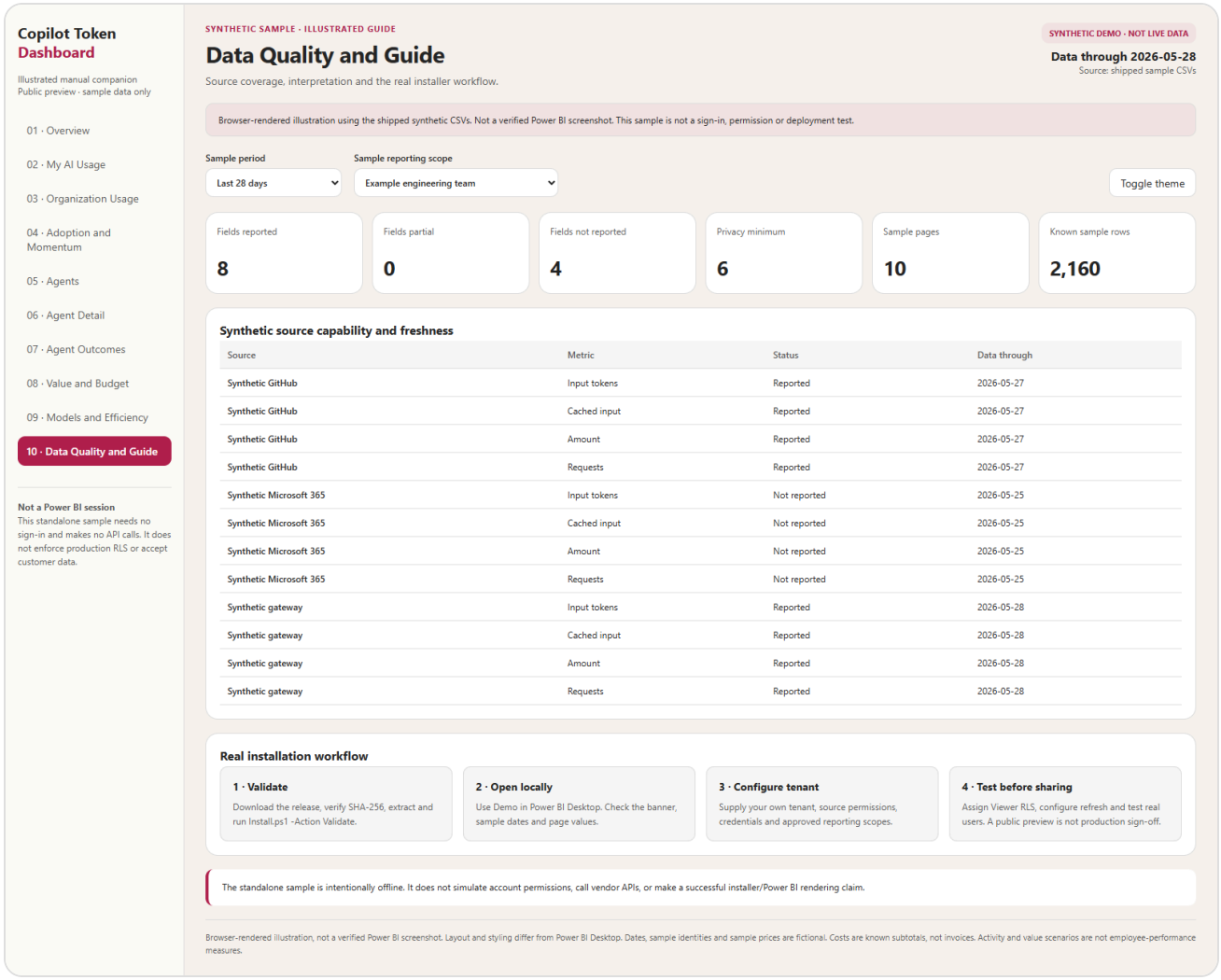


Figure 10. Browser-rendered synthetic source-coverage and installation-workflow illustration, not a verified Power BI screenshot.

The example coverage table has 8 reported, 0 partial and 4 unavailable source-field entries. These are sample capabilities, not a live connection test. Complete Validate, local Desktop checks, tenant/source configuration, Viewer security tests and refresh verification before sharing.

The sample dashboard gallery (SAMPLE_DASHBOARD.md) also explains how to explore the offline HTML and open every original screenshot.